

Top-K youtube Videos ,

Functional requiremnts:

- Top-K videos should be sorted based on the User's activity,
- Top-K videos should get updated after every 5seconds

Non-functional requirements:

- data-partitioning,
- load balancing,
- fault tolerance,
- vertical scalability,
- horizontal scalability,
- low latency,
- high throughput,
- CAP theorem (Here eventual consistency is fine but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:

- user,
- videos,
- history,

API :
POST:/upload
{
 userID,
 {filemetadata}
}
GET:/search/top/{K}

HLD :

